PART A – 20 OBJECTIVE QUESTIONS [20 MARKS]

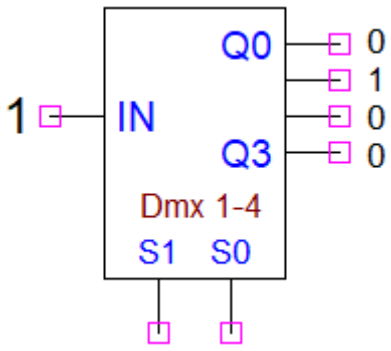
Answer all questions in Appendix A on Page 9

2. Which logic function is used to change a BCD number to its equivalent decimal number?

   A. Decoder
   B. Encoder
   C. Multiplexer
   D. Demultiplexer

3. A multiplexer is a logic circuit that _____.

   A. accepts one input and produces one output
   B. accepts many inputs and produces one output
   C. accepts one input and produces many outputs
   D. accepts many inputs and produces many outputs



5. Which combinational logic function detects error in digital codes during transmission from one point to another in a digital system?

A. multiplexer
B. demultiplexer
C. parity checker
D. code checker

PART B – SUBJECTIVE QUESTIONS [80 MARKS]

QUESTION 1 [15 MARKS]

a. A partially completed truth table for a full adder is given in Table 1. Write the correct values for (i) to (viii). [4m]

Table 1: Partially completed truth table for full adder

Inputs			Outputs	
C_{in}	A	B	C_{out}	Sum
0	0	0	0	(i)
0	0	1	0	(ii)
0	1	0	(iii)	1
0	1	1	(iv)	0
(v)	0	0	0	1
(vi)	0	1	1	0
1	1	(vii)	1	0
1	1	(viii)	1	1

b. Give all possible combination values of A and B in a table which would make output ($A > B$) is true. [4m]

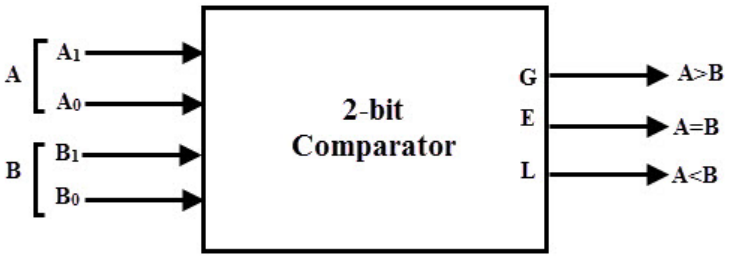


Figure 3: 2-bit comparator

c. One example of encoder application is shown in Figure 4. It is used as a magnetic positional control for ships navigation. In here, the angular position of a compass is converted into a digital code by a 74LS148 8-to-3 line priority encoder and input to the systems computer to produce navigational data output. Magnets and reed switches could be used at each compass point to indicate the needles angular position.

Based on Figure 4, generate the encoder truth table which indicates all angular positional code (Q_0, Q_1 *and* Q_2) for all compass positions (*North, East and etc*). [7m]

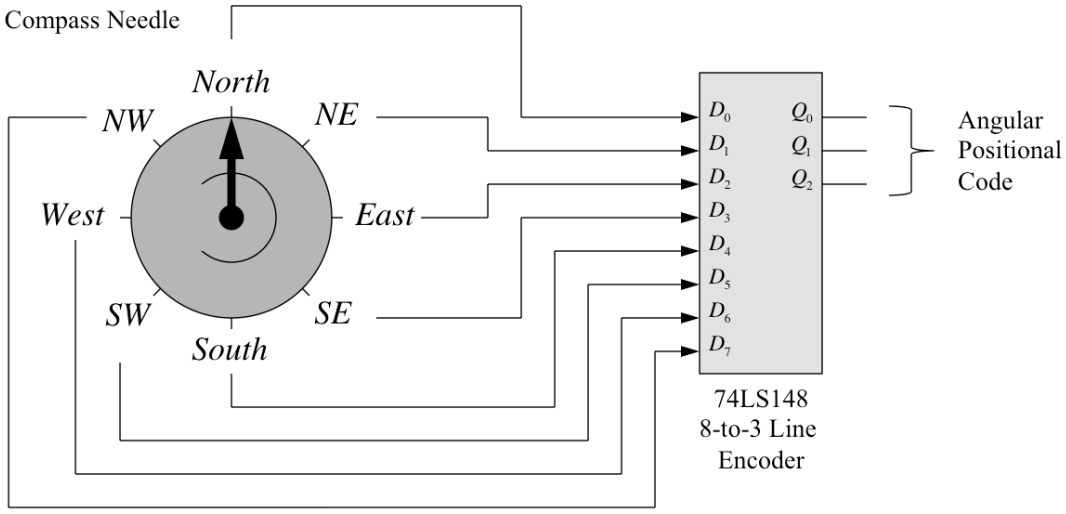


Figure 4: Priority Encoder Navigation